IN THEIR SHOES: EMPATHY THROUGH INFORMATION

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We explore the mechanics of empathy. We show that information about an outgroup can activate and magnify empathy when presented in conjunction with an experience simulating their struggles. This response increases the willingness to help the struggling group. We provide evidence for this effect in an immersive virtual reality experiment where participants (witnesses) experience a simulation of the struggle of unauthorized migrants (protagonists), then replicate these results in a series of controlled lab experiments. We show that information enhances the witnesses' empathetic response and drives them to engage in more prosocial behavior when it increases their perceived interpersonal similarity, or relatability to the protagonist — an effect we trace to attention: eye-tracking data reveal that information provision concentrates witnesses' gaze on the struggles of the protagonist instead of searching through peripheral elements of the scene. Conversely, only information packages that strengthen perceived relatability — an effect that can vary across subgroups with heterogeneous attributes — magnify empathy. Together, our evidence suggests that the ability to put oneself in the shoes of another person or group can be enhanced by activating empathy through simple, targeted, information provision. *JEL codes:* C9, D8, D9, F22, Z1.

I. Introduction

Humans experience hardship, from natural disasters and ethnic conflicts to unsafe or abusive labor conditions. The eventual circumstances of the group experiencing hardship (protagonists) often depend on the response of others who witness those struggles (witnesses). If witnessing hardship breeds empathy, then others may be moved to help through charitable donations, collective mobilization, or voting for policies to support the struggling group. If empathy is absent, then those who are struggling are unlikely

The Quarterly Journal of Economics (2026), 1–32. https://doi.org/10.1093/qje/qjag033.

to get help. How is empathy activated? Why do different people witnessing the same episodes of struggles by others respond differently? Is there scope for interventions or policy to facilitate the ability to put oneself in the shoes of others?

This article explores the mechanics of empathy and tools that can be used to enhance it. We show that information about a struggling group can activate and magnify empathy when presented in conjunction with an experience simulating that group's struggles. This empathetic response increases willingness to help the struggling group, but it is only activated when the information comes before the experience, not after. We first show this in a field experiment where participants witness the plight of migrants crossing the U.S. Southern border in an immersive virtual reality experience. This virtual reality piece, called Carne y Arena® and created by Academy Award® winner director Alejandro González Iñárritu, is the closest to a real-life experience we could hope for.[1] Providing participants with statistical information about unauthorized immigration enhances the empathetic response to the experience—but only if it is provided before and not after it. This is reflected in more positive attitudes and prosocial behavior toward immigrants. We show the robustness and generalizability of our finding in a preregistered lab experiment with a setup that mirrors the Carne y Arena environment: participants are given information about a disadvantaged group either before or after witnessing that group experience hardship. They choose how much to donate to a member of the disadvantaged group. We replicate the results from the field: information presented before simulating the experience of the outgroup activates empathy and increases donation amounts relative to a condition in which the same information is presented after the experience.

We propose a model to shed light on a potential mechanism underlying this magnification of empathy: shifts in attention and in perceptions of interpersonal similarity, or relatability. We develop a framework in which information received before witnessing the struggles of a protagonist may induce the witness to shift attention to attributes of the protagonist they can relate to. Attending to the protagonist's shared features then leads the witness to experience the hardship as if in the protagonist's shoes, which activates empathy and leads to prosocial behavior toward

1. Carne y Arena received a special Academy Award, the only time such an award was given to a virtual reality piece.

the protagonist. We present four lab experiments that provide evidence consistent with the proposed mechanism. Our first study explores the link between information, attention, and the prosocial response. In a novel experimental setup, participants are immersed in a video that simulates the struggles of the urban poor in Pakistan. We use eye-tracking technology to measure attention during the experience. Consistent with our hypothesis, presenting information about the protagonists before the video shifts attention to relatable features of the protagonist, babies and young children in that video, and away from peripheral features that are unlikely to be relatable, which has downstream consequences for prosocial behavior. The second study measures and manipulates relatability directly and shows that it mediates the effect of information on prosocial behavior. The third study highlights that not all information has the capacity to activate empathy: presenting participants with information that does not increase relatability has no downstream effect on prosocial behavior. The fourth study provides further support for the relatability channel by showing predictable heterogeneous treatment effects of information; the same information affects prosocial behavior as a function of the witness's attributes—but only to the extent to which they matter for relatability. Finally, we present additional, suggestive evidence that the same mechanism may operate in observational data.

Before turning to a detailed description of our findings, we note that our analysis concerns instances where information enhances empathy through increased relatability. We acknowledge that a symmetric mechanism may allow information — for example, antimigrant propaganda — to reduce empathy through decreased perceptions of relatability to those suffering hardship. We do not explore such cases in this article.

We begin by building intuition with a simple conceptual framework (Section II). A witness observes a protagonist going through an unpleasant or painful experience. We assume that the witness will have a stronger empathetic response if, when observing the protagonist's struggles, they focus their attention on or learn about attributes of the protagonist that they share and can relate to. This suggests a new channel through which information can shape empathy, in addition to any direct effect on prosocial behavior. If information cues the witness to pay attention to or learn about attributes of the protagonist that overlap with their own — forming a representation that is more relatable — it will change their experience of witnessing the struggles of others; they will experience the struggle as if in their shoes, which magnifies their empathetic response. This simple conceptual framework has two important implications. First, it predicts an asymmetric empathetic response depending on whether information is received before or after the experience of witnessing others' struggles: information acquired before can shift how relatable the struggling protagonist feels to the witness and alter the nature of the experience; information acquired after, which is not top of mind at the time of the experience, cannot alter how the experience is perceived. Second, this framework suggests a way to measure the role of information in shaping the witnessing experience: revealing or shifting attention toward (away from) attributes the witness shares with the protagonist will be accompanied by increases (decreases) in the witness's perceived relatability toward the protagonist. Importantly, the framework also predicts that not all information interventions will be effective in activating empathy — only those that cue relatable categories with overlapping attributes, which implies heterogeneous treatment effects as a function of the witnesses' attributes.

To bring this conceptual framework to the data, we proceed in two steps. In a first step (Section III), we show evidence that information enhances the empathetic response to the experience of witnessing the struggles of the protagonist if it comes before the experience, but not if it comes after — the main prediction from our conceptual framework. In a second step (Sections IV, V and VI), we show evidence that this empathy-enhancing role of information operates by shifting the witness's attention to the painful experience of the protagonist and by increasing how relatable the witness perceives the protagonist to be — the main assumption of our conceptual framework. We describe each step in turn.

In the primary demonstration of the effect, we use a controlled field experiment to recruit participants (the witnesses) who are randomized into two treatments that vary the order of the following: (i) the immersive virtual reality experience Carne y Arena, created by director Alejandro González Iñárritu, which simulates the struggles of unauthorized migrants (the protagonists),[2] and (ii) statistical information on unauthorized immigration in the United States. We measure participants' empathetic response through a targeted charitable donation decision and ques-

2. The immersive museum experience Carne y Arena is described in details in Section III.

tions on immigration policy views. Compared with an untreated control group, attitudes toward immigration improve by 70% (p-value $< .01$) when information precedes Carne y Arena, substantially more than the 36% increase (p-value $< .01$) when information comes after Carne y Arena. The 34% difference between the two treatment effects is large and significant (p-value $< .01$).

Our conceptual framework suggests that the novel effect of information arises because it shifts the witness's attention toward relatable attributes of the protagonist; focusing on these attributes during the experience enhances their ability to connect with the protagonist's circumstances while observing their struggles. To study this proposed mechanism, we design a novel experimental paradigm that emulates the Carne y Arena setting but allows us to directly measure attention through eye-tracking. The experience involves an immersive video about the struggles of Pakistan's urban poor. Using measures of attention derived from eye-tracking data, we show that participants who receive information about the protagonists before the experience focus their attention more on the children and their family members instead of other, presumably less relatable features in depiction. They also donate more money to the protagonists' cause over other charitable options. These results provide evidence for the shift-in-attention-to-relatable-features mechanism in our conceptual framework.

We then design a preregistered conceptual replication of the Carne y Arena study to reproduce the main findings in a more controlled setting. We recruit India-based participants (protagonists), and U.S. and U.K.-based participants (witnesses) who have a chance to receive a bonus payment. The Indian participants are assigned a task both tedious and arduous: counting ones and zeros in large matrices. The U.S. and U.K. participants are told about the work Indian participants were asked to perform and are presented with a series of samples of this task. They are also shown statistical information about Indian citizens, either before or after simulating the experience of the Indian workers by viewing samples of the task. To measure the empathetic response of the U.S. and U.K. participants, we ask them how much of their bonus payment they would want to share with a randomly selected Indian worker. This lab experiment reproduces the finding of our field experiment: U.S. and U.K. participants give 28% more of their bonus payment if presented with information before the experience compared to after (p-value $< .01$).

Having established the validity of this experimental protocol for replicating the effects observed in the field study, we use it to further examine the mechanism. We first look at the hypothesized role of relatability in mediating the effect of information on prosocial behavior. In a similar setting as above, we ask the U.S. and U.K. participants, after they make their donation decisions, "To what extent do you feel like you understand and relate to the circumstances of the Indian workers?" We designed this question based on the psychology literature on interpersonal similarity (relatability) to measure shifts in this construct depending on when information is received (Byrne 1961; Echterhoff, Higgins, and Levine 2009; Schomerus et al. 2025). We use a mediation model (Baron and Kenny 1986) to test the extent to which shifts in relatability can explain the behavioral effect of our treatments. Results show that the effect on donations of receiving information before witnessing the struggles of others is indeed primarily mediated through relatability: information received before the witness's experience increases self-reported relatability, which in turn increases donations; controlling for relatability, the direct effect of information on donations becomes smaller and is no longer significant. This suggests that the treatment effect on donations of information received before simulating the witness's experience operates primarily through an increase in relatability to the protagonist, consistent with our conceptual framework.

Two follow-up experiments using the same paradigm show that not all information about the protagonist's group is sufficient to activate empathy: receiving information that does not increase relatability will also fail to activate empathy and have little downstream consequences for prosocial behavior. We use a data-driven approach to leverage variation in the relatability of information. In the first study, we start by presenting U.S. and U.K. participants with a series of 24 information exhibits and ask them whether they can relate to each. We then select the 12 least relatable information exhibits. These are presented as the information exhibit in the same experimental design described above. Despite keeping the design otherwise unchanged, receiving unrelatable information has no effect on donations compared with receiving no information. In the second study, we investigate the role of heterogeneous attributes in U.S. and U.K. participants, following a similar approach. We start by asking U.S. and U.K. participants to rate the relitability of 24 information exhibits, randomly selected from a pool of 100 exhibits. This allows us to identify four information pieces that women find highly relatable, that men do not find relatable, and for which female and male participants show the greatest divergence in relatability. We provide these four exhibits to a separate set of participants before they witness the arduous task of the Indian workers. Consistent with our model, women perceive the Indian worker as more relatable after receiving this information, but men do not; women's donations are significantly higher compared to no information, while men's donations remain unchanged.

Finally, we show suggestive evidence that a mechanism similar to the one outlined in our conceptual framework also operates in observational data. We study a setting that is similar to our experimental framework, where we analyze charitable donations from donors across U.S. counties, using data extracted from Bursztyn et al. (2024). The protagonists are people in three countries devastated by natural disasters: Haiti, Japan, and the Philippines. The witnesses are residents in U.S. counties. We measure the empathy response to witnessing the suffering in those countries as charitable donations in after these disasters. The observational analogue to the information treatment in our experimental framework is a measure of the likelihood of interpersonal contact with people from those three countries, computed from a new large-scale survey (n = 2,400) across U.S. counties. Motivated by the evidence in Bursztyn et al. (2024) who show that contact with Arab Muslims induces better knowledge of information about Arab Muslims and Islam, we conjecture that a high (low) likelihood of contact is equivalent to an intense (mild) information treatment. To isolate plausibly exogenous variations in contact across county-country pairs, we use quasi-random historical immigration shocks from Burchardi, Chaney, and Hassan (2019). Finally, to measure our hypothesized mediating factor, relatability, survey participants answer a short personality test (McCrae and Costa 1987) and are then incentivized to guess how many personality traits they share with a person from Haiti, Japan, or the Philippines. We conjecture that this measure reflects perceived similarity, or relatability, in the absence of any specific context. We show first that plausibly exogenous variations in contact increase charitable donations to Haiti, Japan, or the Philippines, replicating the finding in Bursztyn et al. (2024) for Arab Muslim countries.[3] We adapt the mediation analysis with instrumental variables from Dippel, Ferrara, and Heblich (2020), and show that contact increases perceived relatability, which in turn increases charitable donations; controlling for both contact and relatability, only relatability has a significant effect on donations. This suggests that the effect of contact on donations operates primarily through an increase in relatability.

I.A. *Related Literature*

This article contributes to several strands of literature. First, it relates to an extensive body of work on intergroup contact and empathy, dating back to Allport (1954) (for meta-analyses see Pettigrew and Tropp 2006; Paluck, Green, and Green 2019; Lowe 2024). A key focus of this literature has been the evaluation of interventions — experimental or quasi-experimental — that facilitate interactions between groups, such as cooperative tasks, school integration, or mixed housing policies (Bazzi et al. 2019; Rao 2019; Mousa 2020; Lowe 2021; Corno, La Ferrara, and Burns 2022; Kaplan, Spenkuch, and Tuttle 2024). This body of work highlights the importance of conditions like equal status and shared goals in achieving positive outcomes, consistent with Allport's original "contact hypothesis."[4] Beyond contact, a number of studies suggest positive effects of perspective-taking interventions, where individuals are encouraged to imagine the world through the point of view of the outgroup (Broockman and Kalla 2016; Adida, Lo, and Platas 2018; Kalla and Broockman 2020, 2023a; Alan, Gumren, and Kubilay 2021; Rodríguez Chatruc and Rozo 2024; Adida et al. 2025; Arman et al. 2025). Siddique, Vlassopoulos, and Zenou (2026) show that information dissemination, through a documentary about education, increases prosociality from a dominant ethnic group in Bangladesh toward ethnic minorities and analyzes facial expressions revealing empathy. We advance this literature by proposing and providing evidence for a mechanism through which contact and perspective-taking may shift attitudes via attention and relatability. In addition, we add to a growing literature on information provision — see the recent

3. With a single foreign-origin group, Bursztyn et al. (2024) cannot control for county fixed effect. With data on three countries and many counties, we are able to control for both county and country fixed effects when predicting contact.

4. Enos (2014), Hangartner et al. (2019), and Lowe (2021) show that contact can backfire in settings where these conditions are not met.

review by Haaland, Roth, and Wohlfart (2023), and in the context of immigration Haaland and Roth (2020) and Alesina, Miano, and Stantcheva (2023)—and bring a new channel through which information, even information not explicitly designed to be persuasive, can change attitudes.

Second, our work contributes to a long-standing literature in social psychology, and more recently in neuroscience, on the role of perceptions of others in building empathy (see Krebs 1975; Davis 1994).[5] Recent studies have focused on lab experiments manipulating labels of in-groups versus out-groups, for example, Vaughn et al. (2018) who examine neural responses to observing pain in others, or Hagenbach and Kranton (2024) who measure whether one subject is able to remember information about shared traits with another, depending on whether they compete or cooperate. We use practical measures of relatability and empathetic responses; consider a commonly-used policy tool (information provision); and study policy-relevant, empathy-inducing events, such as unauthorized migrations or natural disasters. We bring the question to natural settings.

Third, our research contributes to the literature in psychology on how perceptions of interpersonal similarity foster empathy. Byrne (1961) first proposed the similarity-attraction effect, where perceptions of interpersonal similarity engender more positive attitudes toward the target. Work on homophily is consistent with this hypothesis, finding that people with similar traits are more likely to trust and support one another (McPherson, Smith-Lovin, and Cook 2001). For example, people are more likely to express empathy toward others' struggles as perceived interpersonal similarity increases (Wei and Liu 2020). We build on this work, showing that perceptions of interpersonal similarity can be shifted through information provision, which then prompts a greater empathetic response.

Fourth, our findings contribute to the work on how people form mental representations of their environment and how these representations differ from objective features of that environment due to memory and attention constraints. These constraints limit the number of objects a person can attend to and keep in their working memory at any given time (Luck and Vogel 1997; Oberauer et al. 2016). As a result, people form simplified represen-

5. More broadly, our work belongs to a long tradition in economics of modeling altruism (Becker 1974).

tations of the environment that focus on a limited set of features that are either salient at the time of judgment due to bottom-up cues (e.g., visual salience), or are top of mind due to the category that is activated at the time of judgment (Nosofsky, Kruschke, and McKinley 1992; Markman 2013). A stream of research has highlighted the implications of such constraints for representing economically relevant information environments (Loewenstein and Wojtowicz 2023). For example, Ba, Bohren, and Imas (2022) and Bordalo et al. (2024) demonstrate the implications of simplified mental representations for belief updating; Bohren et al. (2024) and Bordalo et al. (2023) highlight the implications for choice and risky decisions. Recent work by Bordalo et al. (2024) incorporates these factors into a formal model of choice, where the decision context cues a mental category that channels attention top down to features of the environment. We build and contribute to this work by showing how a similar mechanism affects people's interpersonal mental representations, and how the overlap between these representations can activate empathy and prompt prosocial behavior.

Fifth, this article relates to research in media and communication studies that aims to explain why and how audiences engage with entertainment narratives. In line with our framework, affective disposition theory links dispositions initially formed toward characters with the emotional reactivity to the subsequent plights of those characters. This ultimately drives the viewer's hedonic response to the resolution of the narrative (Zillman and Cantor 1977; Raney 2004, 2017).

The rest of the article is structured as follows. Section II provides a simple conceptual framework to guide the interpretation of our empirical setup and results. Section III presents evidence for our main finding: information magnifies the empathy response to the experience of witnessing the struggles of an outgroup if it comes before the experience, relative to after — the main prediction from our conceptual framework. Section IV offers evidence that information alters how the witness experiences observing the struggles of the protagonist, by inducing greater attention to relatable features. Section V shows evidence suggesting that information amplifies empathy by increasing perceptions of relatability to the outgroup which, in turn, transforms the experience of witnessing their struggles — the main assumption of our conceptual framework. Section VI provides supportive evidence in observational data. Section VII concludes.

II. Conceptual Framework: The Mechanics of Empathy

We propose a simple model to formalize the interpretation of our empirical results. We are interested in how a specific experience of witnessing the struggles of an outgroup can induce empathy toward that group, and how factors such as information can amplify this empathetic response.

A witness — labeled w — will have to decide whether they want to help a protagonist — labeled p — after they observe p going through an unpleasant or painful experience. Let $G(w)$ correspond to the group that w belongs to, for example, politically liberal Americans, the majority group among visitors to the Carne y Arena exhibit; and $G(p)$ correspond to the group that p belongs to, for example, unauthorized immigrants from Latin America. We propose a framework to characterize the strength of the empathetic response of the witness while observing the protagonist. Following the psychology literature (McPherson, Smith-Lovin, and Cook 2001; Wei and Liu 2020), we conjecture that the empathetic response is more likely to be activated when w views p as similar, or relatable, to themselves. Information before the experience — but not after — can magnify this response if it reveals or shifts attention to attributes of the protagonist's group $G(p)$ that have a greater overlap with those of the witness's group $G(w)$.

The witness's group is characterized by a vector of attributes, $\mathbf{a}(w) = \big(a_1(w), \cdots, a_N(w)\big)$, with $a_n(w) \in \{0, 1\}$. $a_n(w) = 1$ means that group $G(w)$ has attribute n, and $a_n(w) = 0$ that it does not.[6] Similarly, the witness perceives the protagonist's group as characterized by a vector of attributes, $\mathbf{a}(p) = \big(a_1(p), \cdots, a_N(p)\big)$. We follow the literature in cognitive psychology (Luck and Vogel 1997) and let bounds on attention and working memory prevent the witness from considering the entire set of objective attributes at a given time. Instead, they form a simplified mental representation of groups $G(w)$ and $G(p)$ based on the attributes they attend to. Following Nosofsky, Kruschke, and McKinley (1992), we intro-

6. We present our conceptual framework with binary values for attributes, $a_n(w) \in \{0, 1\}$, for simplicity. All results can readily be extended to attributes as a continuous variable in the interval [0,1], where $a_n(w) > a_n(w)$ means that attribute n is more prominent than attribute m. The choice of nonnegative values is without loss of generality, as negative attributes can simply be encoded as their opposite. With finitely many attributes, the choice of the bounded interval [0,1] is also without loss of generality, allowing for any relative prominence.

duce bounded attention across attributes in the form of attribute-specific weights. These attention weights can be driven bottom-up by environmental factors such as salience (Bordalo, Gennaioli, and Shleifer 2013), or top-down by the category $\mathcal{C}$ of attributes that are cued by the decision context (Bordalo et al. 2024). Specifically, let the vector $\boldsymbol{\alpha}(\mathcal{C}) = (\alpha_1(\mathcal{C}), \ldots, \alpha_N(\mathcal{C}))$ correspond to the attention weights that the witness allocates across the attributes given category $\mathcal{C}$, with $\alpha_n(\mathcal{C}) \in [0, 1]$. A greater $\alpha_n(\mathcal{C})$ indicates that more attention is given to attribute n. We define the attention-weighted set of attributes of the witness's group, $\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(w)$, as the element-wise product between the attention vector, $\boldsymbol{\alpha}(\mathcal{C})$, and the attributes vector, $\mathbf{a}(w)$,

$$\text{(1)} \quad \mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(w) \equiv \boldsymbol{\alpha}(\mathcal{C}) \circ \mathbf{a}\,(w) = \big(\alpha_1(\mathcal{C})a_1(w), \cdots, \alpha_N(\mathcal{C})a_N(w)\big)\,.$$

We refer to $\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(w)$ as the witness's mental representation of their own group. Similarly the witness's mental representation of the protagonist's group is $\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(p) \equiv \boldsymbol{\alpha}(\mathcal{C}) \circ \mathbf{a}\,(p)$.

The mental representation, $\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(w)$, corresponds to the set of attributes that come to mind (Gennaioli and Shleifer 2010) when the witness thinks about their own group, and $\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(p)$ corresponds to the attributes that come to mind when they think of the protagonist's group.

We assume that the witness's empathetic response to seeing the struggles of the protagonist depends on perceived interpersonal similarity—relatability—which is captured by the overlap between the witness's mental representation of their own and the protagonist's groups. This overlap increases the ease with which one can put themselves in the shoes of another. Specifically, as w witnesses the struggles of p, it is easier for them to simulate themselves going through the same struggles the more they view p as similar to them—the more w can *relate* to p (Byrne 1961; Wei and Liu 2020). This generates empathy. Formally, empathy induced by observing the protagonist going through an unpleasant experience increases with the dot product of the witness's mental representations of her own group and the protagonist's:

$$\text{(2)} \qquad \mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(w) \cdot \mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}\,(p) = \sum_{n=1}^{N} \big(\alpha_n(\mathcal{C})a_n\,(w)\big) \times \big(\alpha_n(\mathcal{C})a_n\,(p)\big)\,.$$

With binary attributes, the dot product $\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(w) \cdot \mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}\,(p)$ is simply the number of attributes shared between w's representations of the groups, weighted by the attention they pay to those attributes.

Following the literature on empathy (Hoffman 2008), we assume that greater empathy translates into more prosocial behavior toward p.

This conceptual framework provides a simple structure for the mechanics of empathy. Defining $\theta_{w,p,\mathcal{C}} \in [0, \pi/2]$ as the angle, in absolute value, between the two vectors $\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(w)$ and $\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(p)$, we can decompose the strength of the empathetic response into three easily interpretable components,

$$\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(w) \cdot \mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(p) = ||\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(w)|| \times ||\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(p)|| \times \cos\left(\theta_{w,p,\mathcal{C}}\right). \quad (3)$$

First, empathy is stronger the larger $||\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(w)||$, which corresponds to a witness who is intrinsically more universalist (Enke, Rodríguez-Padilla, and Zimmermann 2022), attending to more of their positive attributes, and more likely to feel empathy toward any protagonist. Second, it is stronger the larger $||\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(p)||$, which corresponds to a protagonist who is, given the category $\mathcal{C}$ of attributes cued by environmental factors and the decision context, intrinsically more likable, more likely to receive empathy from any witness. Third, it is stronger the lower the angle $\theta_{w,p,\mathcal{C}}$, which corresponds to a tighter alignment between the witness's mental representations of groups $G(w)$ and $G(p)$.[7]

This conceptual framework allows us to characterize how information that reveals or shifts attention to shared attributes can change the perceived relatability of the protagonist and thus alter the witness's empathy when received before witnessing the struggles of the protagonist. Let $\mathcal{I}$ be the information set available to the witness. With information $\mathcal{I}$, the vector of attributes of the witness's group becomes $\mathbf{a}(w|\mathcal{I})$, that of the protagonist's group becomes $\mathbf{a}(p|\mathcal{I})$, the category of attributes the witness attends to becomes $\mathcal{C}|\mathcal{I}$ and the attention weights become $\boldsymbol{\alpha}(\mathcal{C}|\mathcal{I})$, such that the strength of the empathetic response to observing the protagonist struggle becomes

$$\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C}|\mathcal{I})}(w|\mathcal{I}) \cdot \mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C}|\mathcal{I})}(p|\mathcal{I}) = (\boldsymbol{\alpha}(\mathcal{C}|\mathcal{I}) \circ \mathbf{a}(w|\mathcal{I})) \cdot (\boldsymbol{\alpha}(\mathcal{C}|\mathcal{I}) \circ \mathbf{a}(p|\mathcal{I})). \quad (4)$$

7. Those three components are not fully independent. For instance, $||\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(w)|| = N \Leftrightarrow \alpha_n(\mathcal{C})a_n(w) = 1, \forall n$, and if the perceived attributes of the protagonist are uniformly distributed, then the angle $\theta_{w,p,\mathcal{C}}$ is smaller in expectation than if $||\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(w)|| < N$. We also note that the angle $\theta_{w,p,\mathcal{C}}$ is not defined if $\alpha_n(\mathcal{C})a_n(w) = 0, \forall n$, or if $\alpha_n(\mathcal{C})a_n(p) = 0, \forall n$. But in those knife-edge cases the dot product remains well defined, $\mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(w) \cdot \mathbf{a}_{\boldsymbol{\alpha}(\mathcal{C})}(p) = 0$.

By construction, information received after the experience, not available to the witness when entering this experience, has no effect on the relatability of the protagonist during the experience and thus does not affect the intensity of the witness's empathetic response. The asymmetric effect of information received before versus after the experience is the main prediction of our conceptual framework and our main empirical result (section III).

Our framework posits a mechanism through which information can amplify empathy, relatability, and suggests a simple way to measure relatability. If information alters how the witness perceives their own group or the protagonist's group — a shift in the attributes vectors $\boldsymbol{a}(w|\mathcal{I})$ and $\boldsymbol{a}(p|\mathcal{I})$ — or if it cues a category that places weight on shared attributes — a shift in the attention vector $\boldsymbol{\alpha}(\mathcal{C}|\mathcal{I})$ — then it will tighten the alignment between w's representations of groups $G(w)$ and $G(p)$. This results in a lower angle $\theta_{w,p,\mathcal{C}|\mathcal{I}}$, and increases perceptions of relatability.

As an example, consider the case examined in our first study where (mostly) politically liberal American visitors (w) witness the struggles of unauthorized migrants (p) at the Carne y Arena exhibit. We follow Bordalo et al. (2024) and Evers, Imas, and Kang (2022) in positing that what comes to their mind in their mental representations is at least partly a function of the category $\mathcal{C}$ that is cued by the environment and the decision at hand. For instance, providing information about wildfires in California or about the pope's health can cue the more narrow category of caring about climate change in w's representation of their own group $G(w)$, or of the protagonist being Catholic in w's representation of the outgroup $G(p)$. Such information would shift attention to a category $\mathcal{C}$ of attributes that tends to decrease perceptions of relatability. Alternatively, information can cue a more universal category and channel attention to attributes that are more likely to be shared. For example, information about families in Hispanic communities increases attention to the common attribute of being a parent, which would induce many witnesses to perceive the protagonist as more relatable or more similar to them. In this case, the witness should be more likely to respond positively to the question "To what extent do you feel like you understand and relate to the circumstances of the [protagonist]?", our measure of relatability in the lab experiments of Section V; they should also be more likely to say that they share many personality traits with the protagonist, our second measure of interpersonal similarity in the observational data of Section VI.

The first measure, relatability, quantifies interpersonal similarity directly after a subject witnesses the struggle of a specific protagonist. The second measure, perceived similarity, is more abstract and meant to capture latent interpersonal similarity in the absence of any context. In addition, our framework provides necessary conditions for information to enhance empathy: if information cues categories with little overlap in attributes, then it should not be effective at inducing prosocial behavior.

To conclude, despite having few restrictive features, our simple conceptual framework entails several key testable predictions. First, if information magnifies the empathy response to the experience of witnessing the struggles of the protagonist, this magnification should operate only if information comes before the experience, not after. Second, the main assumption behind this prediction is that the empathy response — measured as an increase in altruistic actions in favor of the protagonist — is due to a shift in attention toward attributes of the protagonist they can relate to , measured as relatability. This process alters the witness' experience when observing the struggles of the protagonist as if in their shoes. This implies that we should be able to see both the shift in attention and relatability as upstream measures of prosocial behavior.

At some level, "feeling like one understands and relates to the circumstances of a [protagonist]" may almost sound like the definition of empathy. However it does not tautologically imply that such stated feelings translate into prosocial behavior or altruistic actions in favor of the protagonist. The empirical results that follow show that when exposed to information about an outgroup before witnessing their struggles, people increase their prosocial behavior, that is, undertake more costly altruistic actions (Section III). Information increases prosocial behavior because it alters the witness's experience when observing the struggles of the protagonist, measured through shifts in attention (Section IV). We also show that this increase in prosocial behavior is primarily mediated through an increase in relatability (Section V.C). Importantly, when information about the outgroup does not cue categories with overlapping attributes (Section V.D), which varies with the heterogeneous attributes of witnesses (Section V.E), this effect disappears.

Before presenting our empirical findings, we discuss some limits of our conceptual framework. First, we recognize that experience and information may also affect empathy directly, for instance through a classical Bayesian updating channel. In that case, the ordering of information and experience should not matter for the strength of the empathetic response they induce. By comparing two experimental treatment arms, one where information comes before the experience and one where it comes after, we can control for any such Bayesian updating channel. Second, we focus solely on the mechanics of empathy during the experience of witnessing the struggles of a protagonist. It is possible that similar mechanics operate before the experience, when subjects imagine a future experience, or after, when subjects remember a past experience. If they do so, then our main result—that information received before an experience induces a stronger empathetic response than information received after—simply requires that these forces operate at a lower intensity after than during the experience. Third, we note that our conceptual framework abstracts from many other relevant features of empathy. For instance, the strength of the empathetic response presumably depends on the intensity of the suffering of the protagonist, or on the talent of the person telling their story. Since neither our experimental protocols nor our observational setting allow us to quantify variations in the emotional intensity of the witness's experience or in the quality of the story telling, we do not explicitly model them. Finally, though we focus on the mechanism where information purely shifts attention across attributes, it may also be possible that the witness learns about the shared attributes and focuses on them during the experience. We cannot distinguish between these two mechanisms in our study, but we view them as complementary.

III. Information Magnifies Empathy: The Carne y Arena Experience

We present our main finding: statistical information about an outgroup magnifies the empathetic response of a person witnessing the struggles experienced by this outgroup if it is presented before the experience, compared to after. We show evidence for this asymmetric effect of information in a controlled in-the-field experiment.[8]

8. AEA registry for randomized control trials (AEARCTR-0009194 on June 8, 2022,) and approval from the University of Chicago Social and Behavioral Sciences Institutional Review Board (IRB22-0551). Our finding on the asymmetric effect of information was not preregistered. We replicate our results in a controlled labora-

III.A. Methods

Our in-the-field experiment features two main treatments, a virtual reality immersive experience treatment — Carne y Arena — where participants witness the struggles of unauthorized migrants crossing the Southern border and being apprehended by border patrol; and an information treatment where participants learn about unauthorized immigration to the United States. Our outcome variable is a measure of attitudes in favor (or against) immigration. By randomly varying the ordering of the Carne y Arena and information treatments, we are able to test whether information, if it comes before, modifies the effect of the Carne y Arena experience on attitudes in favor of immigration. This manipulation is similar in spirit to Gneezy et al. (2020), who vary the timing of information to study motivated reasoning.

Carne y Arena is an award-winning museum-based virtual reality (VR) piece created by Alejandro González Iñárritu.[9] The visitor to the museum is immersed in the experience of unauthorized migrants crossing the U.S. southern border, based on true accounts. The exhibit has three stages. First, the visitor enters a room that is a replica of the cells where unauthorized migrants apprehended at the border are held. They are invited to remove their shoes and wait several minutes. The room is cold and contains artifacts from migrants recovered in the border deserts: backpacks, shoes, water bottles. Second, they enter, barefoot, a large space covered with the same rough sand as the border deserts and are fitted with a VR set. In this VR, they are immersed with a group of unauthorized migrants crossing the border and experience a series of interactive scenes that end with the migrants being apprehended and processed by border patrol. The migrants are tired, one of them is injured, and they are terrified. The visitor can move around the protagonists as if they were there. If they walk "through" a protagonist, they can feel their heartbeat. The VR experience culminates with a final scene where an armed border patrol officer orders the visitor themselves to kneel, pointing his weapon directly at them. Third, having left the VR space and recovered their shoes, the visitor is told the piece was created to reproduce the actual experience of a real group of migrants and

tory experiment that was preregistered (Wharton Credibility Lab registry, AsPredicted no. 204323 on December 12, 2024, https://aspredicted.org/md7z-trrj.pdf), which we present in Section V.B.

9. See a short description and a trailer at https://phi.ca/en/carne-y-arena/.

border patrol officers and is invited to read through their short testimonies. The visit lasts about 15 minutes.

Our information treatment presents participants with statistics about unauthorized immigration to the United States. It consists of a series of 12 exhibits containing information about border crossings (e.g., “In the fiscal year 2020, U.S. Customs and Border Protection apprehended a total of 400,651 people on the southwest border”), about the economic conditions in the migrants’ origin countries (e.g., “The average standard of living in the top four origin countries of migrants apprehended on the southwest border is six times lower than that in the U.S.”), and about their living conditions once in the United States (e.g., “In Texas, unauthorized immigrants are 55% less likely than U.S. born citizens to be arrested for a violent crime”).[10] While our information treatment is not designed to be persuasive or to make all attributes of migrants be viewed more favorably, we conjecture that, as in the mechanism formalized in the model of Section II, it will have a positive effect on relatability because it induces participants to focus their attention on attributes of immigrants they share or perceive as important: caring for education, seeking better economic conditions, fleeing violence, and so on. We confirm in a separate online experiment that the information package we use at Carne y Arena induces participants to perceive migrants as more relatable.[11] It is possible, naturally, that our information treatment may not be perceived as neutral and may induce participants to update their priors, positively or negatively. By comparing the effect on participants exposed to information before versus after Carne y Arena — each receiving the same informational content — we control for such information updating channel.

To measure attitudes toward immigration, we construct an index combining six components. We first ask participants to choose their preferred policies from a list containing two pro-immigration policies — the DREAM Act and asylum policies — and policies un-

10. See Online Appendix B for the full list of information exhibits and the full questionnaire.

11. We follow a procedure similar to the paradigm outlined in Section V.C. We recruited 101 participants on Prolific; 51 received the Carne y Arena information treatment, 50 did not. We ask them “To what extent do you feel like you understand and relate to the circumstances of the undocumented immigrants in the U.S.?” on a 1 (not at all) to 10 (very much so) scale. The information treatment increases this relatability index by 1.37 (p-value = .012), more than one half of a standard deviation (2.62).

related to immigration. Selecting pro-immigration policies reveals positive attitudes (for each, we assign value one if selected, zero otherwise). We ask them to rank their preferred policies and record their ranking of the DREAM Act and asylum policies, if selected: a higher rank for either reveals positive attitudes (we assign a score from one to eight, least to most preferred). We ask participants to choose their preferred policy among anti- and pro-immigration policies: selecting a pro-immigration policy reveals positive attitudes (we assign scores from one for the most anti-immigration policy — deport all unauthorized migrants — to five for the most pro-immigration policy — grant full citizenship to all unauthorized migrants). Finally, we ask participants to choose a charitable donation to be made on their behalf to a charity supporting immigrants, animal welfare, or environmental projects: choosing the immigrant charity reveals positive attitudes (we assign value one if selected, zero otherwise). Each of the six components (support the DREAM Act or not, support asylum policy or not, the rank of the DREAM Act if selected, the rank of asylum policy if selected, immigration policy views, and donate to the immigrant support charity) is individually standardized (mean zero and standard deviation one). Our final index is the standardized sum of those six components.[12] The standardizations are made for the control group so that coefficient estimates are expressed as percentages of a standard deviation in the control group.

Our experimental protocol is designed to measure the effect of the ordered interaction of our information and immersive experience treatments on attitudes in favor of immigration. Participants are randomly assigned to one of four conditions.[13] For the Control condition we measure attitudes before any treatment. For the CyA only condition, we measure attitudes just after participants go through Carne y Arena. For the Info before CyA condition, we measure attitudes after participants have received our information treatment and then gone through Carne y Arena, in that order. For the Info after CyA condition, we measure attitudes after participants have gone through Carne y Arena and then re-

12. Combining outcomes into an index increases precision by decreasing survey measurement error and limits the potential for biases from multiple hypothesis testing (Broockman, Kalla, and Sekhon 2017; Bursztyn, Fujiwara, and Pallais 2017).

13. Our experimental design features additional treatment arms that we do not use in this study. See Online Appendix Table A1.

ceived our information treatment, in that order. We recruit participants who visited the Carne y Arena art installation on site ($n = 718$): at Fair Park in Dallas, Texas (May–June 2022), and at Kaneko in Omaha, Nebraska (June–September 2022). We present results with both locations combined as our baseline and show results for each location separately in the Online Appendix. We keep only data from respondents who reach the end of the survey. The data collection and randomization are done using Qualtrics. The observable characteristics of respondents are balanced between randomized groups, except for gender in some of the smaller Omaha groups (see summary statistics in Online Appendix Table A2, and balance tests in Online Appendix Table A3).

We aim to minimize any form of experimenter demand or Hawthorne effects. All respondents are selected among the same group of museumgoers, all fill out one part of our survey before entering the exhibit and the other part after. Only the ordering of questions varies between treatment arms. Respondents are told that our survey is designed to study "The Power of Art,"[14] which, given that respondents are visitors to an art exhibit, does not reveal the hypotheses we aim to test. We also minimize any form of John Henry effect: only one person at a time is allowed to go through Carne y Arena, so friends cannot communicate about the survey until after they have completed it and exited the exhibit hall; we directly instruct visitors not to communicate with friends or partners about their questionnaire; and respondents answer questions on an individual tablet in a dark, quiet, and secluded waiting area, under a solemn atmosphere, so that they are unlikely to be influenced by others.[15]

Importantly we note that the main hypothesis we test here — information magnifies empathy if it comes before the experience of witnessing the struggles of others compared with if it comes after — was not preregistered. We used the Carne y Arena immersive experience as a unique setting to explore the interaction of

14. We thank Katie Cutright from the Emerson Collective for suggesting this choice of words.

15. The physical setting in Dallas (May–June 2022) and in Omaha (June–September 2022) allowed us to run the before and after sections of our survey without interference: visitors enter and exit the Carne y Arena VR experience in a quiet and dark space inside the exhibit hall. We attempted to run the same experiment in Richmond, CA, but as visitors exited into the crowded space of another exhibit and a restaurant and bar, it proved physically impossible to implement our experimental protocol there.

experience and information. This revealed a novel and unanticipated finding that information can act as a treatment that alters the experience and its effect on empathy. As we discuss later, we reproduce this result in a preregistered laboratory experiment (see Section V.B).

III.B. Main Results

We estimate treatment effects on attitudes toward immigration,

$$Attitudes_i = \alpha + \beta \cdot Treatment_i + \epsilon_i, \tag{5}$$

where $Treatment_i$ takes the value zero or one according to which condition individual i is assigned to, and β measures the effect of a given treatment on attitudes toward immigration. For instance for $Treatment_i = 0$ if $i \in$ Control and $Treatment_i = 1$ if $i \in$ CyA only, β measures the effect of Carne y Arena on attitudes toward immigration, expressed as a percentage of a standard deviation of our attitude index among the control group. We measure attitudes for the control group before they have gone through the immersive experience. Their attitudes therefore correspond to the unconditional attitudes among the selected group of museum goers who chose to visit the Carne y Arena exhibit. For the treatment group (CyA only), we measure attitudes after they went through the exhibit. This population is ex ante identical to the control group, the only difference being their randomized exposure to the immersive experience.

The results are presented in Figure I. We first show that the Carne y Arena experience alone improves attitudes toward immigration by 32% of a standard deviation (top estimate, p-value $<$ .01). Although this result is not the main focus of our article, it is interesting on its own, suggesting that an immersive experience can have a large effect on attitudes.[16] It also confirms that our

16. The effect of Carne y Arena is also persistent. Online Appendix Table A5 shows that attitudes are 41% higher two months after visiting the exhibit compared with before seeing the exhibit. The size and persistence of the effect of the immersive experience of Carne y Arena on attitudes is worth noting. For instance, Kalla and Broockman (2023b), who use a similar measure of attitudes toward immigration, find that the largest and most effective treatment (through discourse rather than VR) moves attitudes by 10%–15% of a standard deviation after 1.5–4.5 months, substantially below the 41% long-run effect of Carne y Arena. Unfortunately, we have too few follow-up respondents to statistically distinguish the long-term effect of alternative orderings of information and Carne y Arena.

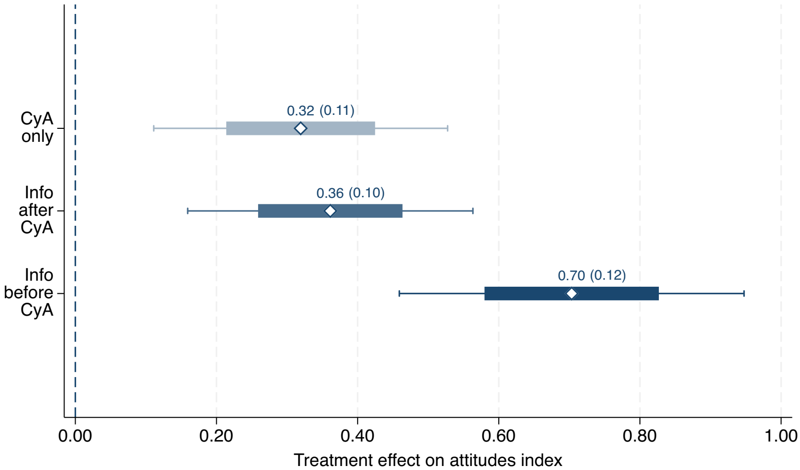


FIGURE I.

Information, Carne y Arena, and Attitudes.

This figure shows the effect of various treatments on attitudes, expressed as a percentage of a standard deviation of attitudes for the control group, β in equation (5). The thick lines represent $\pm$ 1 standard error from the point estimates and the whiskers 95% confidence intervals. Each estimate corresponds to a different regression. The control group is always Control. The treatment groups are, respectively: for the top estimate, CyA only; for the middle, Info after CyA; for the bottom, Info before CyA. Interpretation (Info before CyA, bottom estimate): the attitudes of respondents exposed to the information then Carne y Arena treatments, in that order, are 70% (std. err. 12%) of a standard deviation higher than the attitudes of respondents not exposed to any treatment (control group). See Online Appendix Table A4 for additional statistics.

in-the-field experimental setting is well suited to study the interaction between a striking experience and information.

Our main finding is that the effect of Carne y Arena is magnified if a subject receives our information treatment before Carne y Arena but not after: attitudes improve by 70% of a standard deviation relative to the control group if participants received information before witnessing Carne y Arena (Figure I bottom estimate, p-value $< .01$), substantially more than the 36% increase if participants receive information after witnessing Carne y Arena (Figure I middle estimate, p-value $< .01$). Figure II confirms the asymmetric effect of information and presents our main finding: attitudes improve by 34% of a standard deviation for participants who receive information before Carne y Arena compared to those who receive it after (Figure II, top estimate, p-value $< .01$). This re-

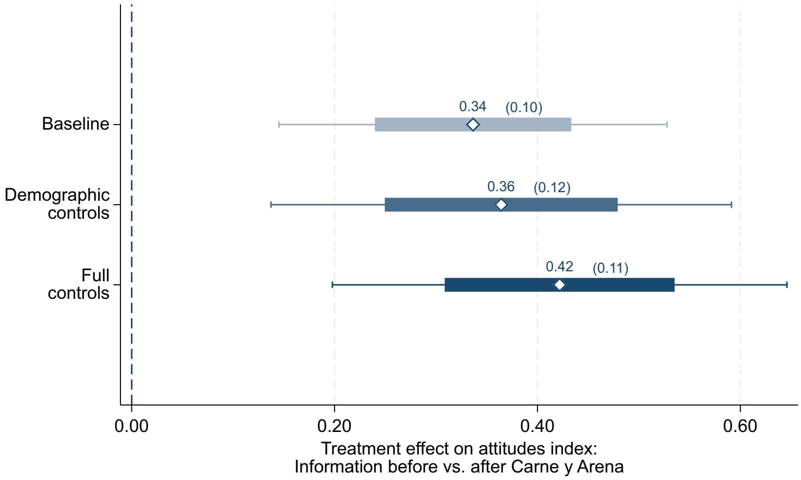


FIGURE II.

Order Effect on Attitudes, Carne y Arena.

This figure shows the treatment effect of receiving information before Carne y Arena, compared to a control group that receives information after Carne y Arena. The thick lines represent ± one1standard error from the point estimates, and the whiskers 95% confidence intervals. The middle and bottom estimates include controls for age, gender, and being born outside the United States. The bottom estimate additionally controls for self-reported strength of emotional response to Carne y Arena. See Online Appendix Table A6 for additional statistics.

sult is robust to including demographic controls and self reported perceptions of emotional reaction to Carne y Arena (Figure II, middle and bottom estimates). Our interpretation is that information shifts people's attention to traits they share with migrants (e.g., the fact that they are parents) and away from peripheral elements of the scene (e.g., the contours of the terrain). This shift in attention induces them to perceive migrants as more relatable and changes the way they experience Carne y Arena, as if they were in the shoes of the protagonists. If instead information comes after, participants 'missed out' on critical aspects of living through the more transformative experience. We provide more direct evidence for this mechanism in Sections IV, V, and VI.

III.C. Additional Results

By comparing the effect of information before versus after witnessing Carne y Arena, both exposed to the same information, we have some confidence that our results are not driven by a separate effect that information updating may have on attitudes, orthogonal to the asymmetric mechanism in the model of Section II. To rule out that participants may pay unequal attention to information depending on when they receive it, which could affect our results, we verify and confirm in Online Appendix Table A7 that participants who receive information before versus after Carne y Arena are equally successful at answering a quiz on the information they received.

Online Appendix Table A8 presents results using only respondents in Dallas or only in Omaha. In Dallas, we can only measure the treatment effect of Carne y Arena and of information after Carne y Arena; they are statistically indistinguishable. In Omaha, we can compare all treatment effects: Carne y Arena is statistically indistinguishable from information after Carne y Arena, and both are statistically smaller than information before Carne y Arena.

Online Appendix Table A9 compares our treatment effects for different types of respondents: Hispanics versus non-Hispanics, foreign versus native born, and liberal versus conservative.

Not surprisingly, all treatment effects are weaker, although with the same rankings, for Hispanics and for foreign born respondents — the majority of whom are of Hispanic origin — presumably because they are more likely to hold attitudes favorable to immigration, which leaves less room for a sizable treatment effect, and because overlapping attributes are already salient, which blunts the effect of Carne y Arena as they may already know personally of the experience of unauthorized migrants, diluting the treatment effects, notably the effect of information on relatability.

More interesting is the comparison of liberal versus conservative respondents. Liberal respondents have a strong reaction to Carne y Arena in all treatments, with a stronger effect if information comes before Carne y Arena. Conservative respondents on the other hand have a weak and insignificant reaction to Carne y Arena alone, or information after Carne y Arena; but they react as strongly as liberal respondents to information before Carne y Arena. This suggests that conservative respondents at baseline perceive unauthorized migrants as less relatable and are therefore less responsive to the Carne y Arena experience; if they receive information before the immersive experience of Carne y Arena, they are induced to relate more to unauthorized migrants, which activates their empathy when they witness their struggles in Carne y Arena. We confirm in a separate online experiment that the information treatment induces only a small and statistically insignificant shift in the perception of relatability to unauthorized migrants among liberal respondents, while it induces a large and significant shift for conservative respondents.[17]

IV. Mechanism: Attention

The previous section presented behavioral evidence for the key prediction of the model of Section II: information activates empathy when it is provided before witnessing the struggles of an outgroup. The mechanism underlying this finding relies on the assumption that information alters how the witness lives through the experience of observing the struggles of a protagonist.

Here we analyze the role information plays in shaping the experience of the witness by measuring attention directly.[18] We incorporate eye-tracking technology to follow the witness's gaze while they are observing the struggles of the protagonist. This allows us to provide evidence for the conjecture that information affects the witness's experience by changing how they attend to relatable features of the protagonist during this experience. A difference in perception, where one group of witnesses is more focused on relatable features of the protagonist than the other, implies that there is a greater capacity for the former to put themselves in the protagonist's shoes — activating empathy.

IV.A. Procedure

We develop a novel experimental design that incorporated the Webgazer eye-tracking protocol into an immersive video. Participants ($n = 601$) are recruited using the Prolific crowd-sourcing platform with the survey being hosted on the Gorilla platform (Anwyl-Irvine et al. 2020).[19] The data were collected in two waves in March 2026, a pilot wave and a preregistered replication. All participants view a video titled "Saving Lives in Karachi," produced by Pakistani journalist Sharmeen Obaid-Chinoy. The video

17. Among 39 liberal respondents, the information treatment increases relatability by 0.94 (p-value = .308). Among 50 conservative participants, it increases by 1.89 (p-value = .018).

18. Preregistrations for the study can be found at AsPredicted no. 280327 (https://aspredicted.org/vs4wq7.pdf).

19. See Online Appendix Table A2 for summary statistics.

is a panoramic experience about a Pakistani pediatrician's efforts to reduce child mortality in the city of Karachi.[20] Most scenes feature children and their families describing the struggles they face in their lives. About half of the scenes take place in a hospital and discuss the poor health outcomes faced by Pakistan's urban poor.

Importantly for our purposes, most scenes have features that are likely relatable, for example, the family and the child receiving care, as well as noncentral features such as hospital equipment and other nonprotagonist characters. Our framework predicts that if information shifts the witness's attention toward features of the protagonist they find relatable, it alters their experience and induces greater empathy. Our goal in this study is to examine whether providing information about the hardships of Pakistan's urban poor families alters the witness's attention during the video, as measured by gaze concentration on babies and young children. If, as we posit, these are relatable features, this will induce the witness to experience the struggles of the protagonist as if in their shoes, with downstream behavioral consequences for prosocial behavior.

We describe the outline of the experiment. First, participants' eye movements are calibrated so that their attention can be tracked through their webcam during the first 10 seconds of the video; see Krajbich, Armel, and Rangel (2010) and Smith and Krajbich (2019) for details on how this is done. Participants are then randomized into two conditions: Info before Experience and Experience only.[21] Demographic information is collected from all of the participants at the beginning of the experiment. For participants in the Info before group, nine pieces of information are provided about economic hardship, especially related to children and infant mortality in Pakistan.[22] All participants then proceed to view the video "Saving Lives in Karachi" with their eye movements tracked through their webcams. Following the video, similar to the Carne y Arena design, participants choose between three

20. See https://www.youtube.com/watch?v=GG7HtmahJzc.

21. The purpose of this study is to test whether information received before the experience of witnessing the struggles of the protagonist shifts the witness's attention to information-related features of the protagonist during the experience, hence the Info before Experience versus Experience only conditions. The asymmetric effect on prosocial behavior of information before versus after the experience that our mechanism implies is analyzed and verified in separate experiments in Section V.

22. See Online Appendix Table A10 for the full list of information exhibits.

charities to donate \$2: The Humane Society of the United States, the Global Rahmah Foundation, and the Natural Resources Defense Council. Only one of the charities is committed to helping improve the livelihoods of Pakistan's urban poor (Global Rahmah Foundation), and we use the choice of that charity as our main behavioral dependent variable indicating prosocial behavior toward the protagonist.

IV.B. Attention Measure

To quantify the participants' attentional responses during the video, we extract gaze metrics from the raw eye-tracking data. Webgazer produces estimates of participants' gaze locations at a rate determined by the participants' webcams, around 30 frames per second if there are no interruptions. This allows us to analyze dynamics of visual attention in response to the video. We prespecify seven regions of interest (ROIs) depicting babies and young children in the video, defined as normalized rectangles in video coordinates.[23] From the estimated gaze coordinates, we measure the total dwell time, that is, the sum of intersample intervals for all gaze samples landing inside each of the seven ROIs.

This metric of total dwell time allows us to assess whether information alters participants' attentional patterns. Higher values indicate that participants focus their attention more on relatable features — babies and young children. By examining treatment effects, we are able to evaluate whether information provision reshapes attention and alters how the witness perceives the experience.

To the best of our knowledge, this is the first time that visual attention is directly quantified to study the type of hypothesis that we propose; this required the development of a new paradigm and empirical design. This new design offers an objective, process-level measure that does not rely on self-reporting. We hope that future research builds on and extends this paradigm to answer a broader set of economic questions.

IV.C. Main Results

First we examine whether the information treatment affects patterns of visual attention. To do so, we focus on the total dwell

23. See Online Appendix Figure A1 for an example of a selected region of interest featuring a young child.

TABLE I.
EFFECTS OF INFORMATION ON GAZE-BASED ATTENTION AND DONATIONS

Metric	Info before Experience	Experience only	Difference	Cohen's d
Total dwell time (ms)	10,049	8,566	+1,483**	0.161
Donation (%)	63.1	47.0	+16.1***	0.328
Observations	301	300		

Notes. Metrics are computed from Webgazer eye-tracking data and from charitable donation choices. "Difference" is Info before Experience minus Experience only. Statistical significance is based on two-sided t-tests using OLS estimates with robust standard errors. Cohen's d expresses this difference as a share of the pooled standard deviation of the attention metrics in each group. $*p < .01$, $**p < .05$, $***p < .01$.

time on relatable features — babies and children — and compare participants in the Info before Experience ($n = 301$) and the Experience only ($n = 300$) conditions.

The attention metrics across both conditions are presented in the top row of Table I.

We observe a significant shift in attention to relatable features as a function of information. Participants in the Info before Experience condition are consistently and significantly more focused on regions of the video that depict babies and children. The effect of information on attention is quantitatively large: 16% of a standard deviation (Cohen's d) more time spent looking at babies and young children.[24]

Our model predicts that greater attention to relatable features of the protagonist fosters empathy. To link the treatment to prosocial behavior, we relate information to donation decisions in the second row of Table I. Receiving information prior to the video significantly increases the probability of donating to the target charity by 16 percentage points. The magnitude of this effect (33% of a standard deviation) is similar to the effect in the Carne y Arena study (38% of a standard deviation, the difference between the Info before CyA and CyA only treatment effects on attitudes in Figure I).

24. The preregistered effect on the pooled regions of interest is consistent across all seven regions individually, all in positive direction — see Online Appendix Figure A2. The effect replicates across each wave of data collection: Cohen's $d = 0.201$ in the pilot wave, $d = 0.192$ in the first replication wave, and $d = 0.162$ in the second replication wave. The effect is similar for the replication waves only ($n = 405$): difference $= \pm$ 1,514 ms, $p = .081$, $d = 0.177$.

These findings provide direct evidence for the mechanism posited in the model of Section II to explain the main result observed in the field — the effect of information in the immersive Carne y Arena experience. Specifically, information that induces the witness to focus their attention on protagonist-relevant features, caring for children in this study, alters the witness's experience of observing the protagonist's struggles and fosters greater prosocial behaviors.

V. Mechanism: Relatability

In the previous section, we provided initial evidence for the model in Section II: information before the experience shifts attention and changes how a witness lives through the experience of observing the struggles of a protagonist. Here we test the downstream consequences of such attentional shifts on relatability.

To do so, we introduce an experimental paradigm that conceptually reproduces the Carne y Arena field experiment. This controlled laboratory setting allows us to experimentally test the cognitive mechanism of our conceptual framework, something that was not feasible in the field. We first replicate the main result of the Carne y Arena study — the asymmetric effect of information when received before versus after witnessing the struggles of the outgroup—in a preregistered experiment.[25] Using the same paradigm, we introduce an empirical measure of perceived overlapping attributes between the witness and the protagonist, which we label relatability, and show that the observed increase in empathy is mediated through changes in relatability. Finally, we explore a richer set of information treatments. Specifically, we show that unrelatable information does not magnify empathy and that information has heterogeneous treatment effects as a function of the witness's attributes.

V.A. Methods

1. Participants.. We recruit participants from two different online subject pools: one from India on CloudResearch and the other from the United States and United Kingdom on Prolific. The first group is asked to complete a tedious and arduous task for

25. Preregistrations for the studies can be found at AsPredicted no. 204323 (https://aspredicted.org/md7z-trrj.pdf), no. 204985 (https://aspredicted.org/wmdg-6bwj.pdf), and no. 212694 (https://aspredicted.org/3my9-j8b9.pdf).

over an hour and be paid a flat fee.[26] Specifically, 100 workers are recruited to each complete 60 effort tasks. One task involves counting the number of zeros in a large, randomly generated table of zeros and ones. This paradigm was used in Falk and Kosfeld (2006), Abeler et al. (2011), and Imas, Kuhn, and Mironova (2022), among others, as a costly effort task to estimate labor supply decisions. Pretesting showed that each table took about one minute to complete. Workers see one table at a time and could not proceed to the next one before entering the correct answer.

Our primary studies concern the behavior of the group from the United States and United Kingdom recruited from Prolific. Participants earn a base fee above the minimum wage of the respective countries (study payment is set at approximately $20/hour); importantly this wage is substantially greater than the wage paid to the Indian workers ($7/hour). Each could potentially earn an additional bonus in the form of a $100 Amazon gift card through a lottery.[27] After learning their own payment information, participants are given a brief description of the task given to the Indian workers and the amount they stand to earn — which is substantially less than their own.

2. Randomization across Conditions.. U.S. and U.K. participants were randomized into one of four potential conditions:[28] Control, Info before Experience, Info after Experience, or Experience only. Those conditions are designed to mimic the same conditions as in the Carne y Arena study.

For the experience component, participants go through a simulation of the Indian workers' experience: each is presented with 10 of the tasks on separate pages. They are also asked to imagine the experience of having to complete the tasks themselves but, importantly, are not required to do so. The experience component is analogous to the immersive virtual reality experience in the Carne y Arena study: participants witness the hardship that the protag-

26. The flat fee is substantially above the prevailing average wage in the country. Study payment is approximately $7/hour compared to a minimum wage of $2–$6 per day, depending on the region.

27. Each Prolific participant is entered into a lottery for the prospect of winning a $100 gift cards.

28. The exact number of conditions varies across studies. We describe in detail the conditions used for each study below.

onist goes through, they partially experience the arduous task, but they are not fully exposed to the same hardship themselves.

For the information component, participants are presented with a set of information exhibits about social and economic conditions in India. We vary the information set across studies. In Sections V.B and V.C, it consists of 12 statements about conditions in India and Indian residents, similar to the information package provided in the field at Carne y Arena.[29] Our aim is to conceptually replicate the main results in the Carne y Arena study, and to test in this conceptual replication whether the magnification of empathy by information is mediated through relatability. In Section V.D, we preselect the 12 least relatable information exhibits from a list of 24 statements. In Section V.E, we explore heterogeneous treatment effects by preselecting four statements that are perceived as relatable as a function of the witness's own features. We explain in detail how we select information and describe the information selected for each study below.

3. Prosocial Behavior.. Participants report their willingness to donate part of their potential earnings to a randomly selected Indian worker. They answer the following question by moving a slider between 0 and 100: "If you win the $100 gift card, we will pair you with a randomly selected Indian worker who was recruited to complete the task. You can use some of the $100 gift card to increase the compensation of the Indian worker in the form of a bonus. How much of the $100 gift card would you be willing to give to the worker?" This measures prosocial behavior toward the Indian worker, analogous to our measure of attitudes in favor of immigration in the Carne y Arena study.

29. The information treatment is designed to mimic the same treatment as in the Carne y Arena study and contains 12 statistical information statements about the social and economic conditions in India, sourced from the World Bank and Gallup (see Online Appendix D for the full list of information exhibits). Statements include "According to the 2021 World Risk Poll, nearly one in four (23%) Indians were 'very worried' that the water they drink could cause them serious harm," "In 2022, 85% of Indian population felt that children in the country have the chance to learn and grow every day," and "India's young women are just as optimistic about their local job prospects as men of the same age." As in the Carne y Arena study, these statements are meant to induce participants to focus their attention on attributes of Indian workers they can relate to — caring about health, education, gender equality — without overtly attempting to be persuasive.

4. Relatability.. To test the conjecture of empathy through shared attributes, each participant is asked to answer a question on the relatability of the Indian workers. They are asked to indicate "To what extent do you feel like you understand and relate to the circumstances of the Indian workers?" on a 1 (not at all) to 10 (very much so) scale. This serves as our measure of the participant's perceptions of shared attributes with the Indian worker—relatability.[30]

Finally, participants answer questions about their demographics and perceptions of the task (how onerous it is).[31] Summary statistics are in Online Appendix Table A2.

V.B. Conceptual Replication of the Carne y Arena Results

We first use our paradigm to replicate the main behavioral finding in Carne y Arena.

Participants are randomly assigned one of two conditions, Info before Experience where they are presented with the 12 information exhibits before going through a simulation of the Indian workers' experience, and Info after Experience, where information comes after the simulation of the experience.

V.B.1. Results.. First, we examine the number of quiz questions answered correctly to ensure that participants in both conditions spent the same amount of effort attending to the information.[32] There are no significant differences between the Info before Experience and Info after Experience conditions (average correct of 8.68 versus 8.83 questions, respectively; p-value for the difference = .70).

Next, we estimate the effect of the treatment manipulation on prosocial behavior (donation amount) using the same specification as in the Carne y Arena study:

$$Donation_i = \alpha + \beta \cdot Treatment_i + \epsilon_i, \tag{6}$$

30. We use the concept of relatability (McPherson, Smith-Lovin, and Cook 2001; Wei and Liu 2020) rather than the related concept of perspective-taking (Adida, Lo, and Platas 2018; Alan, Gumren, and Kubilay 2021) because it offers a more familiar wording for our question.

31. See Online Appendix C for the full questionnaire.

32. Note that the timing of the multiple-choice quiz was identical for all participants—after both the information and the experience exhibits had been presented, in whichever order their condition assigned.

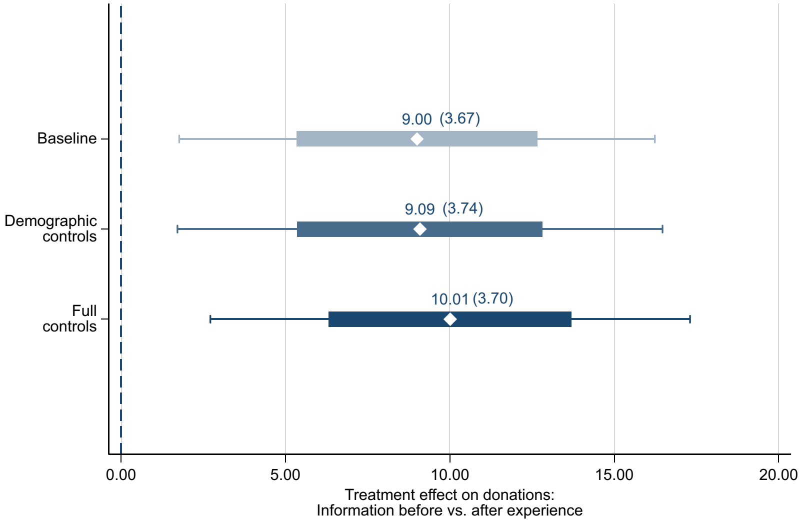


FIGURE III.

Order Effect on Donations, Conceptual Replication.s.

This figure shows the treatment effect of receiving information before the experience, compared to a control group which receives information after the experience, corresponding to various specifications of equation (6). The dependent variable is the amount allocated to the Indian worker. The top estimate has the treatment variable only; the middle estimate includes demographic controls of age, gender, foreign born; the bottom estimate also controls for task perception. The thick lines represent $\pm$ 1 standard error from the point estimates, and the whiskers 95% confidence intervals. See Online Appendix Table A11 for additional statistics.

where $Treatment_i = 1$ corresponds to Info before Experience and $Treatment_i = 0$ to Info after Experience, that is, donation amounts are regressed on whether information came before or after the simulation of the Indian workers' experience. Results are presented in Figure III. Replicating the main Carne y Arena results (Figure II), participants are substantially and significantly more prosocial toward the disadvantaged group if the information comes before the experience than after; participants who receive the information before are willing to donate significantly more than those who receive information after ($9.00 more than the baseline of $33.98, p-value $= .015$). The effect is robust to including demographic controls and to controlling for perceptions of the penibility of the task. This replicates the Carne y Arena result and provides further evidence that information activates the capacity for empathy.[33]

V.C. The Activation of Empathy Is Mediated through Relatability

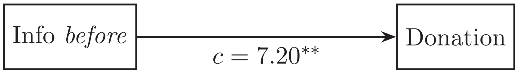
Building on the paradigm of Section V.B, we proceed to test the hypothesis from our conceptual framework (Section II) that the activation of empathy is mediated through an increase in the perception of overlapping attributes between the witness and the protagonist—relatability.

We use the same 12 exhibits for the information treatment as in the study above. Participants are randomized into one of three conditions: Control, Experience only, and Info before Experience. Importantly, after making their donation decisions but before answering demographic questions, we measure each participant's perception of shared attributes with the Indian worker—our relatability measure from the answer to the question "To what extent do you feel like you understand and relate to the circumstances of the Indian workers?" on a 1 (not at all) to 10 (very much so) scale.

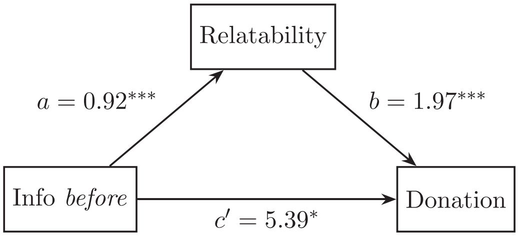
V.C.1. Results.. This experiment allows us to test whether changes in perceptions of relatability drive the shift in donation behavior. We do this by estimating a mediation model (Baron and Kenny 1986) that decomposes the unconditional effect of introducing information before the simulated experience into the indirect effect that acts through changes in relatability and the direct effect of the information treatment, conditional on changes in relatability.

The results of the mediation model are presented in Figure IV. Panel A presents the total—unconditional—effect of information received before witnessing the Indian workers' experience on donation behaviors. Panel B decomposes this total effect into the indirect effect that operates through changes in relatability, versus the direct effect of the Info before Experience treatment conditional on changes in relatability. The mediation model compares the Info before Experience condition to the Experience only

33. We also replicate the Carne y Arena findings in Online Appendix figure A3—the increase in prosocial behavior relative to the Control group is similar for the Experience only and the Info after Experience groups and are significantly smaller than for the Info before Experience group.



(A) TOTAL EFFECT MODEL (c)



(B) MEDIATION MODEL: DIRECT EFFECT (c') AND INDIRECT EFFECT ($a \times b$)

FIGURE IV.

Information, Relatability, and Donations.

Panel A (total effect): information before experience has a positive and significant direct effect (c) on donations, without accounting for relatability. Panel B (mediation model): this panel decomposes the total effect on donations of receiving information before the experience into a direct effect, and an indirect effect mediated through an increase in relatability. Information before experience has a positive and significant effect on relatability (a). Relatability has a positive and significant relationship with donation behavior (b). The direct effect of information before experience (c') is only marginally significant when relatability is included in the regression. All comparisons use OLS regressions, comparing the Info before Experience condition to the Experience only condition. See Online Appendix Table A13 for additional details. Results are similar comparing the Info before Experience condition to the Control condition (see Online Appendix Figure A4 and the corresponding Online Appendix Table A14). $*p < .1$, $**p < .05$, $***p < .01$.

condition. The same model estimated with respect to the Control condition is presented in Online Appendix Figure A4, with similar results.

In Panel A, we see that, in line with the study in Section V.B, providing information about the disadvantaged group before the simulated experience has a substantial and significant effect on donation behavior (arrow c). This total effect is decomposed in Panel B. Consistent with our hypothesis, and the conceptual framework of Section II, providing information before the experience increases relatability (arrow a). Arrows b and c' show that relatability mediates the impact of adding the information treatment on donations. Including both in a regression, relatability has a large and significant impact on donations (arrow b), whereas the direct effect of information becomes smaller and only marginally significant ($p = .095$, arrow c').

These results provide supporting evidence for the mechanism postulated in our model: introducing information before the experience changes the perception of overlapping versus nonoverlapping attributes with the disadvantaged group, measured as increased relatability, which activates the empathetic response to witnessing their struggles and increases the willingness to help.

V.D. Unrelatable Information Does Not Magnify Empathy

Our conceptual framework in Section II proposes a mechanism where information about an outgroup shifts the perception of shared attributes with that outgroup and enhances empathetic responses. For this to occur, the information the witness receives has to highlight features—characteristics or values—they share with the outgroup. Information per se is not sufficient to activate the capacity for empathy: only information that makes the outgroup more relatable can do so.

We take a data-driven approach to test this conjecture in a separate study. A group of U.S. and U.K. participants ($n = 148$) is recruited to rate the relatability of 24 statements about conditions in India and about Indian residents. After being presented with each statement, the participants answer "To what extent do you find this statement relatable?" on a 10-point scale. We take the statements with below-median scores and create a set of 12 unrelatable information statements.[34] Some of the unrelatable statements present India as a competitor to the United States and the United Kingdo, for example, "India's focus on infrastructure and economic expansion is reflected in its 2023 investment levels accounting for 33.32% of GDP, outpacing the U.S. (22.3%) and the U.K. (17.0%)" or "In 2024, India has emerged as the world's fifth-largest economy, surpassing the United Kingdom." Others emphasize specific cultural differences, for example, "Indians have a more circular concept of time than U.S. and U.K. citizens. They are less focused on deadlines and efficiency, and more focused on maintaining relationships" or "Indians view change and progress with suspicion. Progress is viewed as illusory and there is a tight link to the past."

34. All 24 statements are listed in Online Appendix Table A15, with the 12 unrelatable statements we select at the bottom.

We then run an Unrelatable Info before Experience treatment, which mirrors the Info before Experience condition, except that the relatability-enhancing information from the previous studies is replaced with the unrelatable information set. A separate group ($n = 150$) is recruited for this test.[35]

We first verify that the Unrelatable Info before Experience treatment condition does not affect relatability in the same way as the baseline Info before Experience treatment condition does in our previous experiment. This is indeed the case: relatability scores in response to the same "To what extent do you feel like you understand and relate to the circumstances of the Indian workers?" question are significantly lower in the Unrelatable Info before Experience treatment than in the baseline Info before Experience (mean = 5.20 versus 5.91, $p = .027$). Importantly, donation rates are also significantly lower (mean = \$34.28 versus \$42.74, $p < .01$). Donation rates in the Unrelatable Info before Experience condition are similar in magnitude and not statistically different from the Control and Experience only conditions (mean = \$33.55 and \$35.16, respectively). These results provide further support for the proposed relatability-based information channel mechanism described in Section II.

V.E. Heterogeneous Treatment Effects of Information by Group Attributes

Our conceptual framework implies that information will have heterogeneous treatment effects on empathy depending on whether groups of witnesses share the salient attributes of the protagonist. The activation of empathy through information is not universal. Rather, it depends on the congruence between a specific information set and a specific group: only information that makes the outgroup more relatable for a specific group will activate empathy for that group.

We again take a data-driven approach to test this conjecture. We focus on gender to separate participants into two similar-sized groups, women and men. Our goal is to identify gender-specific information exhibits, that is information about the protagonist that is perceived as relatable by women but not by men. We first conduct a survey with 251 Prolific participants based in the United States and United Kingdom. Each participant sees 24 statements

35. The results in this section are robust to controlling for differences in observables between this group and the sample from Section V.C.

about Indian citizens randomly selected from a pool of 100 statements and is asked to rate how relatable they found each statement. As in the preceding study, participants are asked "To what extent do you find this statement relatable?" and respond using a 10-point scale. We perform an analysis of ratings by gender and select the subset of statements such that (i) the mean relatability rating for women is above 5.2, (ii) the mean rating for men is below 4.8, (iii) the difference in mean relatability is at least 1.6 points higher for women than for men, and (iv) the gender difference in relatability is statistically significant (p-value $< .05$). We select four out of six statements that meet those criteria, and a two-tailed t-test reports significant differences in relatability between women and men when filtering for these four statements (5.69 and 3.73, respectively, p-value $= .000$).[36] All four selected female-relatable statements are about the condition of women in India, and they emphasize hardships that women in the United States and the United Kingdom may also face, for example, "30% of women have faced physical violence since age 15" or "Only 0.6% of women report having had a breast-cancer screening."

We recruit 800 new Prolific participants, randomly assigned to two conditions: Experience only ($n = 401$) and female-relatable Info before Experience ($n = 399$; 202 women, 197 men). We substitute the information package of Sections V.B and V.C with the four female-relatable statements. By comparing the treatment effects of information separately for women and for men, we control for any systematic difference in empathy between women and men.

In line with the predicted heterogeneous treatment effect of information, women in the female-relatable Info before Experience treatment are willing to donate significantly more than women in the Experience only condition (\$41.16 versus \$33.07, $p < .01$), whereas the same statements does not significantly affect male donations in the two treatments (\$30.01 versus \$33.07, $p = .279$). The interaction between gender and condition is also significant ($p < .01$). These results show that the proposed relatability-based information mechanism will have heterogeneous treatment effects that are group-specific. These heterogeneous effects may explain why, despite being exposed to information about the hur-

36. All 100 statements are listed in Online Appendix Tables A16–A19, with the four selected statements at the top. The statements are sorted in decreasing relatability for women, with the mean relatability for women and men in columns (1) and (2).

dles of many outgroups, people may not feel empathetic to all of them, but instead allocate their empathy unevenly.

VI. Observational Evidence

We present suggestive evidence that the mechanism we propose and verify experimentally—namely, that feeling similar to and able to relate to an outgroup strengthens empathetic response to witnessing their struggles—also operates in observational data.

We use four observational measures that are broadly analogous to those in our experimental studies: (i) witnessing natural disasters hitting specific foreign countries (Haiti, Japan, and the Philippines), analogous to witnessing the struggles of unauthorized migrants crossing the border, the hardships of Pakistan's urban poor, or the tedious task performed by Indian workers; (ii) personal contact with friends, neighbors, or co-workers of Haitian, Japanese, or Filipino origin, analogous to receiving information about outgroups in our experimental treatments; (iii) perceived similarity towards Haiti, Japan, and the Philippines, analogous to the notion of relatability to Indian workers; and (iv) charitable donations sent to disaster stricken countries, analogous to our measure of attitudes in favor of immigration, donations to a charity helping people living below the poverty line in Pakistan, or sharing a bonus with Indian workers.

We measure the effect of the interaction of contact with a given foreign origin country f, and a natural disaster striking f, on charitable donations toward that country. We hypothesize that residents in a domestic county d where they are more likely to be in contact with people of origin f, feel a stronger sense of similarity to people in country f. Consequently, when country f is struck by a natural disaster, they relate more to the struggles of people in f and are therefore more likely to send charitable donations to help them. Contact with people from f plays a role analogous to information about country f in our experimental studies.

We note that in our observational setting, there is no equivalent notion of receiving information after a disaster strikes country f, unlike in our experimental studies. Instead, because we use variation in contact induced by slow-moving historical immigration shocks, we postulate that contact predates natural disasters, which corresponds to the Info before treatment in our experiments. Our predicted variations in the likelihood of preexisting contacts with people from f, which induces continuous variations in perceived similarity toward f, is therefore the continuous analogue to the binary Info before treatment, and the relatability it induces in our experimental setting.

VI.A. Data

We extract data on natural disasters hitting Haiti (2010 earthquake and cholera epidemic), Japan (2011 Tohoku earthquake), and the Philippines (2013 Bohol earthquake and supertyphoon Yolanda); and on charitable donations toward those three countries by 55,152 U.S. individual donors over 2010–17 from Bursztyn et al. (2024). To measure perceived similarity and personal contact, we conduct a large-scale survey ($n = 2{,}400$) of the U.S. population using Prolific.[37] We introduce an incentivized measure of the perceived similarity between a subject and a person from Haiti, Japan, or the Philippines. This measure is analogous to the notion of relatability, but is more appropriate in a survey where Haiti, Japan, or the Philippines are out of any context. Each respondent first answers a short personality test (big five traits, McCrae and Costa 1987). They are told that we asked the same questions to three people, from Haiti, Japan, and the Philippines. We ask them to guess how many personality traits they share with each, offering financial incentives to form the correct guess.[38] We are not interested in whether a respondent is right or wrong, but only whether they perceive people from foreign origins as similar to them on a personal level. The incentive is solely meant to ensure that they are careful in their answers.

For each country, we quantify perceived similarity as the number of personality traits in common, normalized to mean zero and standard deviation one.[39] We then measure contact with

37. See Online Appendix E for the complete survey. We drop respondents we cannot match to a U.S. county of residence, who do not complete the survey, or are younger than 18 (0.4% of observations). Summary statistics are shown in Online Appendix Table A2. Our resulting sample is somewhat more feminine, foreign-born, Hispanic, and liberal than the U.S. population (see Online Appendix Table A20).

38. We use the answers from a randomly selected respondent from Haiti, Japan, or the Philippines.

39. We show in Online Appendix F that this measure captures plausible variations. For instance, conservative respondents perceive all foreign origins as less similar to them compared with liberal respondents, with a more pronounced differ-

Haiti, Japan, or the Philippines as having a friend, neighbor, or co-worker from those origins, as Bursztyn et al. (2024) do for Arab Muslims. Finally, we remove donations to foreign country f if the likely ancestral origin of a donor is f, based on their name as in Bursztyn et al. (2024), and we remove responses on contact and perceived similarity toward f if a respondent is born in f or has at least one parent born in f.[40] $Donations_{d,f}$ is the inverse hyperbolic sine of the number of charitable donations from U.S. domestic county d to foreign country f.[41] $Contact_{i,d,f}$ equals one if individual i, residing in county d, has contact with people from country f, and zero otherwise. $Similarity_{i,d,f}$ is the normalized index of perceived similarity between individual i, residing in county d, and country f.

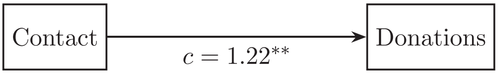
VI.B. Identification

Because there are almost no donations in the absence of a natural disaster and donations flow only after a disaster strikes (see Online Appendix Figure A5), a cross-sectional regression of donations from domestic county d to foreign country f is similar to a difference-in-difference estimate: donations from d to f are approximately the same as the difference between donations after a disaster minus donations before — approximately zero. To identify the causal effect of the interaction of contact and natural disasters, we need plausibly exogenous county-country variations in this interaction term. Natural disasters striking country f are inherently random acts of nature. To isolate plausibly exogenous variations in contact between residents in county d and country f, we control for county and country fixed effects, and use variation in the ethnic composition of U.S. counties induced by quasi-random historical immigration shocks from country f to county d (Burchardi, Chaney, and Hassan 2019). Finally, to decompose the effect of contact on donations into a direct effect and

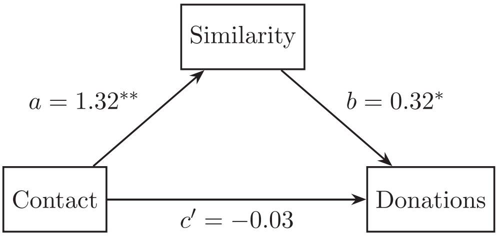
ence for Haiti and the Philippines, which all respondents perceive as less similar than Japan.

40. Contrary to the donation data, where we do not have access to the donor list and must rely on names to infer ancestral origin, we could ask respondents in our survey on similarity about their direct family connections to country f.

41. The inverse hyperbolic sine (or *arcsinh*) function approximates the log function but is well defined at zero: $IHS(x) = \ln\left(x + \sqrt{x^2+1}\right)$. It is commonly used instead of the log function in applied settings with count data that sometimes takes the value zero. It offers an imperfect solution (Chen and Roth 2023) to the known selection biases arising from selectively dropping zeros (Silva and Tenreyro 2006).



(A) Total Effect Model (c)



(B) Mediation Model: Direct Effect (c') and Indirect Effect ($a \times b$)

Figure V.

Contact, Perceived Similarity, and Donations.

This figure quantifies the effect of contact between residents in U.S. domestic county d and foreign country f (Haiti, Japan, or the Philippines) on the inverse hyperbolic sine of charitable donations from d to f when f is struck by a natural disaster (Panel A), and decomposes the total effect into a direct effect, and an indirect effect mediated through an increase in perceived similarity between residents in county d and country f (Panel B). We adapt the mediation analysis with instrumental variables from Dippel, Ferrara, and Heblich (2020). c is the total effect of contact on donations, where contact is instrumented by historical immigration shocks (Burchardi, Chaney, and Hassan 2019), controlling for country and county fixed effects. a is the effect of contact on perceived similarity, where contact is instrumented by historical immigration shocks, controlling for country and county fixed effects. b and c' are respectively the effect of similarity and contact on donations controlling for country and county fixed effects, where similarity is instrumented by historical immigration shocks, treating contact and country and county fixed effects as controls in the first stage. See Online Appendix Table A21 for additional details and Online Appendix Table A22 for results with a single instrument. $*p < .1$, $**p < .05$, $***p < .01$.

an indirect effect mediated through perceived similarity, we adapt the mediation analysis with instrumental variables from Dippel, Ferrara, and Heblich (2020), similar to the analysis of Figure IV.

VI.C. Results

The results are presented in Figure V. Panel A shows that the total effect of contact between residents in U.S. domestic county d and foreign country f on donations from d to f is large and significant (arrow c). Panel B decomposes this total effect into an indirect effect mediated by perceived similarity and a direct effect of contact on donations, conditional on perceived similarity. Contact has a large and significant effect on the measure of similarity (arrow a), which in turns has a large and somewhat marginally significant impact on donations (arrow b). Controlling for the indirect effect mediated by perceived similarity, the direct effect of contact on donations is almost nil and statistically insignificant (arrow c').[42] This suggests that the effect of contact on donations is mediated by similarity, a result that mirrors our experimental evidence showing the impact of information on donations is mediated by relatability (Figure IV).

VII. Conclusion

Using a series of field and lab experiments, and observational data, we study the mechanics of empathy. We show that a person can have a stronger empathetic response after witnessing the struggles of an outgroup if they have been presented with information about that group before witnessing their struggles, compared with after. Importantly, we show that information activates empathy in our data because, and only when, it increases relatability to the outgroup. Being able to relate to the circumstances of others enables a vicarious experience of their struggles — as if one were in their shoes.

In a controlled field experiment, individuals exposed to statistical information about unauthorized immigrants to the United States before an immersive VR experience depicting the hardships faced by unauthorized migrants crossing the border have a stronger empathetic response than if the same information is provided after. They are more likely to donate to charities helping migrants and express more positive political attitudes toward immigration. We conducted an experiment that explicitly measures attention in a similar immersive experience, finding that information prior to the experience shifts attention to features relevant to the protagonist in a manner consistent with our framework. We use a series of controlled lab experiments to identify the proposed

42. With multiple instruments, the decomposition of the total effect into a direct effect of contact on donations, and an indirect effect mediated through similarity does not algebraically sum to 100%. Online Appendix Table A22 shows specifications with a single instrument, immigration shocks in 2000 or 2010, where this algebraic decomposition holds. In both cases, approximately 100% of the effect of contact on donations is mediated through similarity, whereas the direct effect is almost nil as in Figure V. Unfortunately, neither instrument is strong on its own.

mechanism. First, we reproduce our main finding from the field: participants previously exposed to statistical information about India have a stronger empathetic response to witnessing Indian workers performing an arduous task than participants who receive the same information later. We also show that the same information package increases respondents' relatability to the circumstances of Indian workers, and that this increased relatability explains their stronger empathetic response. In contrast, information that does not induce greater relatability fails to enhance empathetic responses; information has heterogeneous treatment effects on empathy as a function of the witness's group identify and attributes. In observational data, residents in counties where they are more likely to be in contact with specific foreign-origin groups (from Haiti, Japan, or the Philippines) feel more similar to these groups and have a stronger empathetic response to witnessing those foreign origins devastated by natural disasters: they send more charitable donations to those foreign countries.

Together, our results suggest a novel mechanism through which political and private attitudes can change: information provision and intergroup contact can improve a person's ability to put themselves in the shoes of others. In particular, we show that beyond a possible direct informational effect, receiving information about an outgroup can shift attention to shared attributes, increasing perceived relatability, and, in turn, enhance empathy—even among those who are initially more hostile toward the outgroup.

Data Availability

The data and code underlying this article is available in Andries et al. (2026), in the Harvard Dataverse, https://doi.org/10.7910/DVN/2BVMKX.

Supplementary Material

An Online Appendix for this article can be found at *The Quarterly Journal of Economics* online.

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